CAN OUR PRICE MODEL WORK IN A CITY IT HAS NEVER SEEN?

Airbnb price analytics on Databricks
Trained on London and Bristol, tested on Greater Manchester

Alina Bodnar
Danylo Kuryliak
Zakhar Veresniuk
Sofiia Vitchynkina

The short answer

The pipeline handled Manchester with no new logic; the model transferred only partly

£100.84

baseline MAE in Manchester

£76.93

model MAE, London + Bristol test

£79.68

model MAE in Manchester, first run

WHAT

In Manchester the model is still 21% more accurate than the baseline.

SO WHAT

But it overprices a typical listing by about £32: it learned each city's price level from coordinates.

NOW WHAT

A new client needs a small labelled price sample, not new code.

One pipeline for any city

Bronze

RAW FILES

Original listings.csv.gz per city and snapshot. SHA-256 checked, never overwritten.

Silver

CLEANED LISTINGS

34.9% of London and 22.9% of Bristol listings have no price, almost only where the calendar is closed. We flag them: scored, never evaluated.

Gold

FEATURES, PREDICTIONS, VIEWS

Price quotes and revenue fields rebuild the price on every paired row, so they are excluded.

Gold tables: features, city_metrics, room_type_metrics, neighbourhood_metrics

ONE NOTEBOOK, CITY AS A PARAMETER

02_etl_pipeline · city = london

02_etl_pipeline · city = bristol

02_etl_pipeline · city = greater-manchester

The model cuts error by 26% against the baseline on 12,434 held-out listings

The model beats the baseline in both cities; Bristol's error falls by 43%



Test-set MAE, GBP per night, 12,434 held-out listings. Source: model results, v1.

BASELINE

Training median price for the same room type × number of guests.

MODEL

Gradient boosting on log price. Features: property and room type, capacity, rooms, coordinates, min/max nights.

HONEST TEST

20% of each city held out before any model choice, used once.

Manchester ran end to end, but the typical error rose by a third

In Manchester the mean error rose 4%, but the typical error rose 34%



Model error per listing, GBP per night. A/B: 12,434 test listings; Manchester: 6,692. Source: City C results.

PIPELINE

One failure: os.link() is not permitted on a Databricks Volume. Fixed with copy plus checksum; cleaning and features untouched.

MODEL (FROZEN V1)

100% of 7,799 listings scored, 6,692 with a price evaluated. MAE £79.68 against £100.84 for the baseline.

The model overprices 78% of Manchester listings because it learned price levels from coordinates

The bigger the listing, the more the model overprices it: from +£13 for one guest to +£132 for seven or more



Median predicted minus actual price, GBP. Manchester first run. Grey band: London and Bristol stay within ±£11.

ONE DIRECTION

Median error +£31.80 in Manchester, against +£4.64 in London and +£1.41 in Bristol.

CAUSE

Every Manchester listing lies 1.65° (≈180 km) north of the training data. A tree model gives it the price level of the northern edge of London and Bristol.

Replacing coordinates made every city worse, so we kept the frozen model

Distance to centre is worse in every city; Bristol's error more than doubles



5-fold cross-validated MAE on London and Bristol training rows. GBP. Source: City C results.

MODEL CHANGES

Distance to centre: rejected by CV
Equal city weights: rejected by CV
Broader property types: rejected
City-relative price: needs Manchester prices

Every dashboard page states a finding; Genie answers from verified queries

DASHBOARD: THE TITLE IS THE FINDING

CITY OVERVIEW

“London has 31 times more listings than Bristol, and its median night costs £175 against £130”

LONDON VS BRISTOL

“Bristol is about a quarter cheaper than London and stays open roughly 50 more days a year”

MODEL

“On unseen Greater Manchester the unchanged model scores £79.68 MAE … 21% better than the £100.84 baseline”

GENIE: WHAT MADE IT ANSWER CORRECTLY

10 verified SQL examples titled with the questions themselves. Genie reuses them and answers the assignment questions right on the first try.

Views with described columns keep it from confusing price, availability and model error.

Defined terms: asked for occupancy, it explains that the data has availability only.

A new city needs a small labelled price sample, not new code

What

The pipeline runs for any city with one parameter. In Manchester the model still beats the baseline by 21%.

So what

It knows a city's price level only through coordinates it has seen, so it overprices a typical new-city listing by about £32.

Now what

Before quoting prices in a new city, collect a labelled price sample or a price index and rescale. Report median error next to MAE.

Four parts, four hard problems

Danylo Kuryliak

EDA AND DATA QUALITY

Built: EDA, leakage review, cleaning rules.
Hard part: A third of London prices missing; quote fields copying the price.

Alina Bodnar

ARCHITECTURE AND ETL

Built: Medallion lakehouse, one parameterised pipeline.
Hard part: Bronze step failing on a Databricks Volume in the Manchester run.

Sofiia Vitchynkina

PRICE MODEL

Built: Baseline, model, test evaluation, City C analysis.
Hard part: Explaining the Manchester gap without using Manchester prices.

Zakhar Veresniuk

DASHBOARD AND GENIE

Built: Gold views, three-page dashboard, Genie Agent.
Hard part: Genie mixing up price, availability and model error.

github.com/DanyloKuryliak/airbnb-price-analytics